

CSE 231 - HOMEWORK I

Problem 1 .:

- 3.4** Given the circuit in Fig. 3.53, calculate the currents i_1 through i_4 .

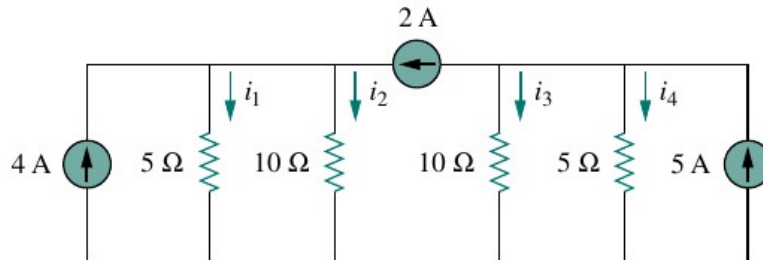


Figure 3.53 For Prob. 3.4.

Problem 2 :

- 3.15** Determine voltages v_1 through v_3 in the circuit of Fig. 3.63 using nodal analysis.

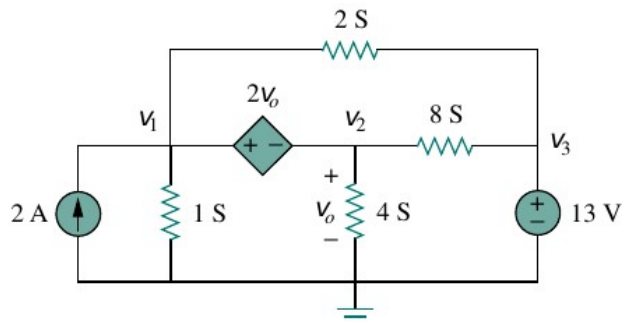


Figure 3.63 For Prob. 3.15.

Problem 3:

- 3.19** Determine v_1 and v_2 in the circuit in Fig. 3.67.

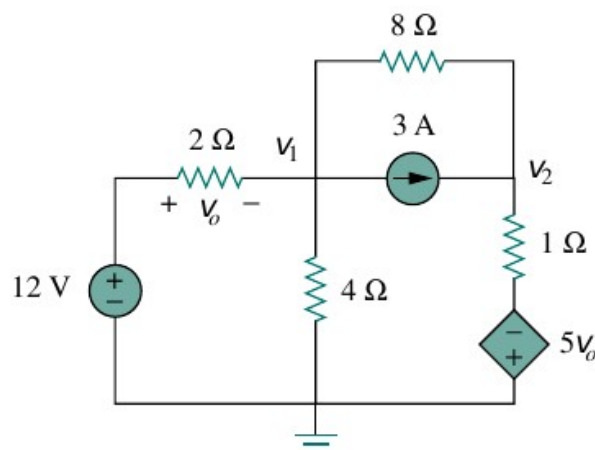


Figure 3.67 For Prob. 3.19.

Problem 4 :

3.21 Find v_o and i_o in the circuit in Fig. 3.69.

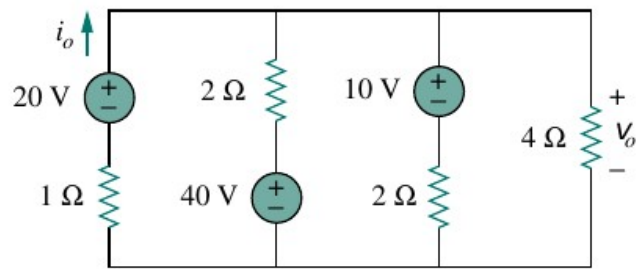


Figure 3.69 For Prob. 3.21.

Problem 5 :

***3.22** Use nodal analysis to determine voltages v_1 , v_2 , and v_3 in the circuit in Fig. 3.70.

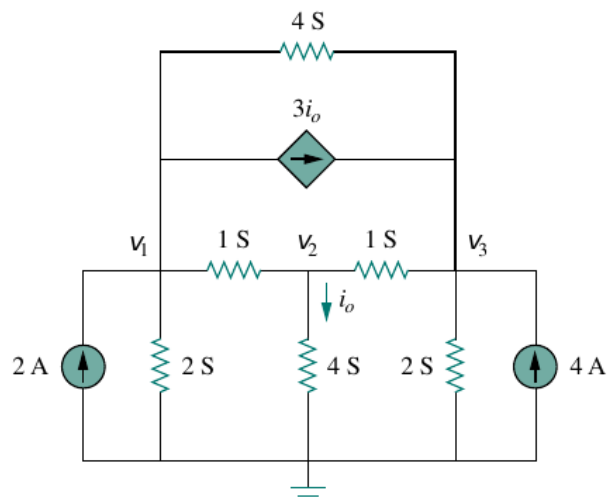


Figure 3.70 For Prob. 3.22.

Problem 6:

3.23 Using nodal analysis, find v_o and i_o in the circuit of Fig. 3.71.

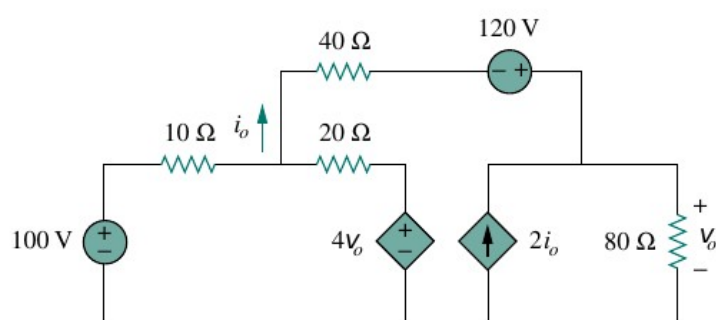


Figure 3.71 For Prob. 3.23.

Problem 7 :

3.24 Find the node voltages for the circuit in Fig. 3.72.

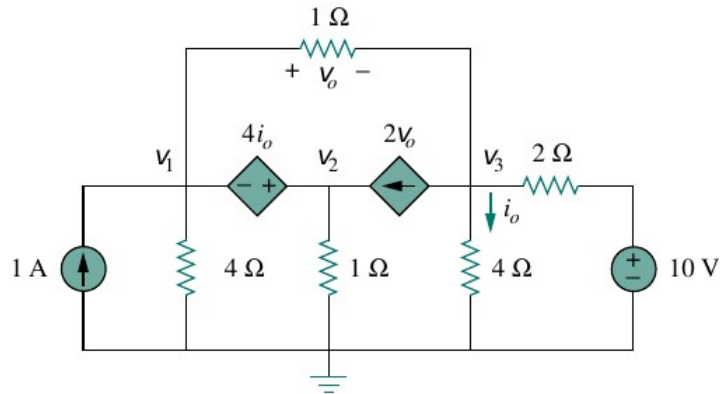


Figure 3.72 For Prob. 3.24.

Problem 8 :

3.33 Apply mesh analysis to find i in Fig. 3.77.

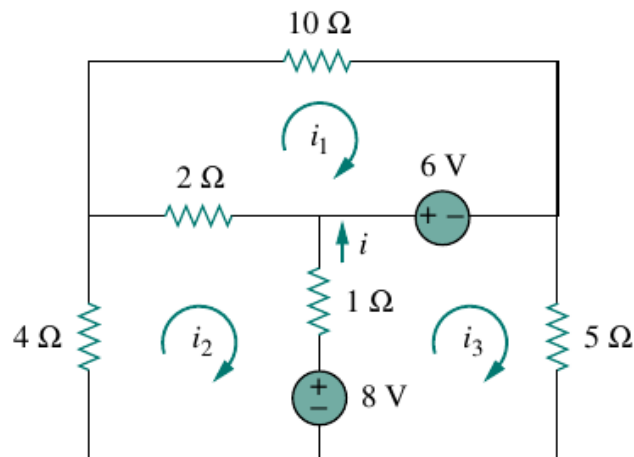


Figure 3.77 For Prob. 3.33.

Problem 9 :

3.44 Find i_1 , i_2 , and i_3 in the circuit in Fig. 3.87.

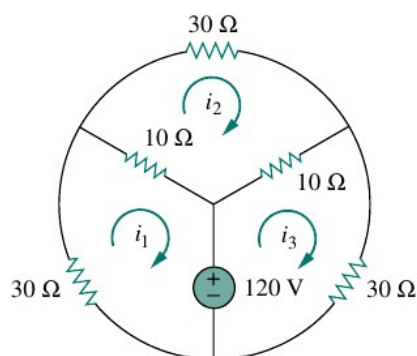


Figure 3.87 For Prob. 3.44.

Problem 10:

- 3.58** By inspection, obtain the mesh-current equations for the circuit in Fig. 3.100.

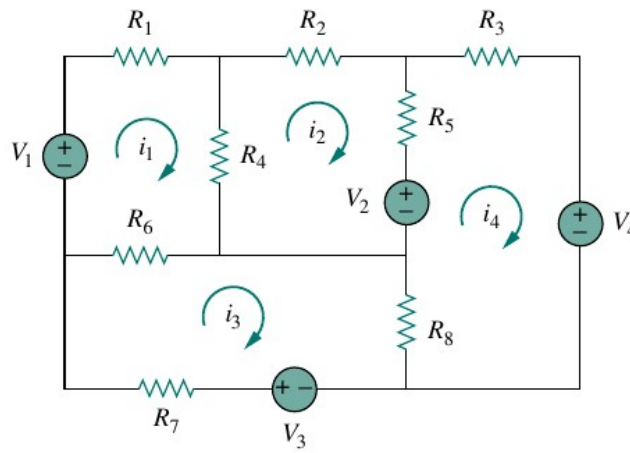


Figure 3.100 For Prob. 3.58.

Problem 11 :

- 3.57** Write the mesh-current equations for the circuit in Fig. 3.99.

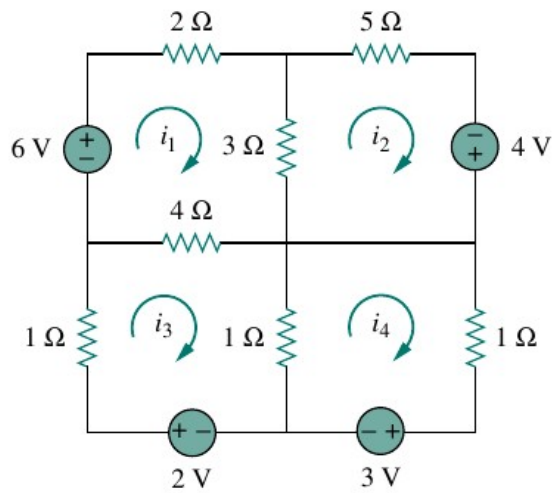


Figure 3.99 For Prob. 3.57.